

Marco Aurelio Nuño Maganda

Profesor Investigador en Ciencias de la Computación

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Fecha de Nacimiento: **12 de Agosto de 1978**

Lugar de Nacimiento: **Mexico, D. F.**

Nacionalidad: **Mexicana**

Teléfono: 834-147-28-35

ORCID: <https://orcid.org/0000-0003-0102-8227>

Scopus: <https://www.scopus.com/authid/detail.uri?authorId=15846253000>

Sci Profile: <https://sciprofiles.com/profile/449039>

Google Scholar: <https://scholar.google.com/citations?user=hSfqx4EAAAJ&hl=en>

Research Gate: https://www.researchgate.net/profile/Marco_Nuno

Publons: <https://publons.com/researcher/1603078/marco-aurelio-nuno-maganda>

ARTICULOS PUBLICADOS EN REVISTAS ARBITRADAS E INDEXADAS

- [27] M. E. García-Luna, S. Polanco-Martagón, Y. Hernández-Mier y **M. A. Nuño-Maganda**, “Comparative Study of Phonocardiogram Classification Using Time–Frequency Transforms and Convolutional Neural Networks,” en *Artificial Intelligence – COMIA 2026*, H. Ponce, H. J. Escalante y R. A. Vazquez, eds. Cham: Springer Nature Switzerland, oct. de 2026, págs. 611-624, ISBN: 978-3-032-37444-8. DOI: [10.1007/978-3-032-37444-8_45](https://doi.org/10.1007/978-3-032-37444-8_45).
- [26] S. Polanco-Martagón, Y. Hernández-Mier, **M. A. Nuño-Maganda** y R. Posada-Gómez, “Automated small-scale hydroponics: a low-cost fuzzy-IoT system for non-specialist users,” *Open Agriculture*, vol. 11, n.º 1, pág. 20 250 532, oct. de 2026. DOI: [10.1515/opag-2025-0532](https://doi.org/10.1515/opag-2025-0532).
- [25] R. M. Cobos-Ramos, **M. A. Nuño-Maganda**, E. Rocha-Rangel, Y. Hernández-Mier y J. A. Castillo-Robles, “An Augmented Reality and Interactive 3D Mobile Visualization of Multifunctional Hybrid Materials: A Prototype and a Preliminary User Experience Evaluation,” *Virtual Worlds*, vol. 5, n.º 4, oct. de 2026. DOI: [10.3390/virtualworlds5040053](https://doi.org/10.3390/virtualworlds5040053).
- [24] C. A. Flores-Gámez, V. I. García-Cervantes, J. A. Gonzalez-Perales, A. Gonzalez-Tovar y **M. A. Nuño-Maganda**, “Developing a Computer Vision-Based Handwritten Recognition, Modeling, and Simulation of Deterministic and Nondeterministic Finite Automata: A Multidisciplinary Approach,” en *Data Analysis Techniques and Applications*, L. C. Méndez-González, I. J. González-Hernández y M. I. Rodríguez Borbón, eds. Cham: Springer Nature Switzerland, mayo de 2026, págs. 95-112, ISBN: 978-3-032-13703-6. DOI: [10.1007/978-3-032-13703-6_4](https://doi.org/10.1007/978-3-032-13703-6_4).
- [23] **M. A. Nuño-Maganda**, K. M. Saldaña-Martínez, C. Zavala-López, Y. Hernández-Mier y S. Polanco-Martagón, “Design and Validation of a Multidisciplinary Mobile App Simulator of a Real-Time Human Body Modeling and Movement Tracking,” en *Data Analysis Techniques and Applications*, L. C. Méndez-González, I. J. González-Hernández y M. I. Rodríguez Borbón, eds. Cham: Springer Nature Switzerland, mayo de 2026, págs. 469-480, ISBN: 978-3-032-13703-6. DOI: [10.1007/978-3-032-13703-6_18](https://doi.org/10.1007/978-3-032-13703-6_18).
- [22] S. Cadena-Flores, J. H. Barrón-Zambrano, J. M. Torres-Arce, J. d. J. Rangel-Magdaleno y **M. A. Nuño-Maganda**, “From Data to Diagnosis: A Systematic Review of EEG Databases for Epilepsy,” *Archives of Computational Methods in Engineering*, feb. de 2026, ISSN: 1886-1784. DOI: [10.1007/s11831-026-10521-x](https://doi.org/10.1007/s11831-026-10521-x).
- [21] J. H. Barrón-Zambrano, **M. A. Nuño-Maganda**, M. Hernández-Díaz, J. de Jesús Rangel-Magdaleno e Y. Hernández-Mier, “Mobile2D-3D-RoboticSim: A robotic platform for computational thinking assessment in STEM and STEAM education,” *SoftwareX*, vol. 33, pág. 102 473, feb. de 2026, ISSN: 2352-7110. DOI: [10.1016/j.softx.2025.102473](https://doi.org/10.1016/j.softx.2025.102473).
- [20] Y. Hernández-Mier, **M. A. Nuño-Maganda**, S. Polanco-Martagón, Á. D. Cantú-Castro, R. Posada-Gómez y J. H. Barrón-Zambrano, “Machine Learning and Queuing Algorithm Integration for Real-Time Citrus Size Classification on an Industrial Sorting Machine,” *Processes*, vol. 14, n.º 1, ene. de 2026, ISSN: 2227-9717. DOI: [10.3390/pr14010164](https://doi.org/10.3390/pr14010164).
- [19] S. Polanco-Martagón, Y. Hernández-Mier, **M. A. Nuño-Maganda**, J. H. Barrón-Zambrano, A. Magadán-Salazar y C. A. Medellín-Vergara, “Comparison of Deep Neural Networks for the Classification of Adventitious Lung Sounds,” *Journal of Clinical Medicine*, vol. 14, n.º 20, oct. de 2025, ISSN: 2077-0383. DOI: [10.3390/jcm14207427](https://doi.org/10.3390/jcm14207427).
- [18] **M. A. Nuño-Maganda**, G. Amador-Cano, Y. Hernández-Mier, Y. V. Saenz-Aguilar y S. Polanco-Martagón, “Automatic Detection and Counting of Nannochloropsis oceanica Microalgae: A Comparative Study of Classical Machine Learning Versus Deep Learning–Based Approaches,” *Aquaculture, Fish and Fisheries*, vol. 5, n.º 6, e70129, oct. de 2025. DOI: [10.1002/aff2.70129](https://doi.org/10.1002/aff2.70129).

- [17] Y. Hernández-Mier, **M. A. Nuño-Maganda**, S. Polanco-Martagón, G. Acosta-Villarreal y R. Posada-Gómez, "Unsupervised Optical Mark Recognition on Answer Sheets for Massive Printed Multiple-Choice Tests," *Journal of Imaging*, vol. 11, n.º 9, sep. de 2025, ISSN: 2313-433X. DOI: [10.3390/jimaging11090308](https://doi.org/10.3390/jimaging11090308).
- [16] **M. A. Nuño-Maganda**, Y. Hernández-Mier, S. Polanco-Martagón, J. H. Barrón-Zambrano, J. C. Elizondo-Leal y A. Díaz-Manríquez, "Low-Cost Versatile Remote Healthcare Monitoring of Bedridden Patients," en *Smart Devices for Medical 4.0 Technologies*, M. Guduri, C. Chakraborty y M. Margala, eds., **ISSB: 9781003603610**, Boca Raton: CRC Press, abr. de 2025, cap. 2, págs. 18-41. DOI: [10.1201/9781003603610-2](https://doi.org/10.1201/9781003603610-2).
- [15] J. F. López-Luna, E. N. Armendáriz-Mireles, **M. A. Nuño-Maganda**, H. Herrera-Rivas, R. Machucho-Cadena y J. A. Hernández-Almazán, "Design and validation of a preliminary instrument to contextualize interactions through information technologies of health professionals," *Health Informatics Journal*, vol. 30, n.º 2, 2024. DOI: [10.1177/14604582241259323](https://doi.org/10.1177/14604582241259323).
- [14] **M. A. Nuño-Maganda**, I. A. Dávila-Rodríguez, Y. Hernández-Mier, J. H. Barrón-Zambrano, J. C. Elizondo-Leal, A. Díaz-Manríquez y S. Polanco-Martagón, "Real-Time Embedded Vision System for Online Monitoring and Sorting of Citrus Fruits," *Electronics*, vol. 12, n.º 18, sep. de 2023, ISSN: 2079-9292. DOI: [10.3390/electronics12183891](https://doi.org/10.3390/electronics12183891).
- [13] C. A. Calles-Arriaga, J. L. Hernández, M. Hernández-Ordoñez, **M. A. Nuño-Maganda**, M. B. O. Moctezuma y E. Rocha-Rangel, "Magnetic Field Computer Interface Based on Cartesian Robot for Practical Demonstration of Ampere's Law in Precollege Courses," *Physics Education*, jul. de 2022, [10.1088/1361-6552/ac8518](https://doi.org/10.1088/1361-6552/ac8518).
- [12] C. A. Rocha-Herrera, A. Díaz-Manríquez, J. H. Barron-Zambrano, J. C. Elizondo-Leal, V. P. Saldivar-Alonso, J. R. Martínez-Angulo, **M. A. Nuño-Maganda** y S. Polanco-Martagón, "EEG Feature Extraction Using Evolutionary Algorithms for Brain-Computer Interface Development," *Computational Intelligence and Neuroscience*, vol. 2022, pág. 7571208, jun. de 2022, ISSN: 1687-5265. DOI: [10.1155/2022/7571208](https://doi.org/10.1155/2022/7571208).
- [11] **M. A. Nuño-Maganda**, J. H. Jiménez-Arteaga, J. H. Barron-Zambrano, Y. Hernández-Mier, J. C. Elizondo-Leal, A. Díaz-Manríquez, C. Torres-Huitzil y S. Polanco-Martagón, "Implementation and integration of image processing blocks in a real-time bottle classification system," *Scientific Reports*, vol. 12, n.º 1, pág. 4868, mar. de 2022, ISSN: 2045-2322. DOI: [10.1038/s41598-022-08777-x](https://doi.org/10.1038/s41598-022-08777-x).
- [10] J. C. Elizondo-Leal, J. G. Ramirez-Torres, J. H. Barrón-Zambrano, A. Díaz-Manríquez, **M. A. Nuño-Maganda** y V. P. Saldivar-Alonso, "Parallel Raster Scan for Euclidean Distance Transform," *Symmetry*, vol. 12, n.º 11, nov. de 2020, <https://doi.org/10.3390/sym12111808>, ISSN: 2073-8994. DOI: [10.3390/sym12111808](https://doi.org/10.3390/sym12111808). dirección: <https://www.mdpi.com/2073-8994/12/11/1808>.
- [9] M. Hernández-Ordoñez, **M. A. Nuño-Maganda**, C. A. Calles-Arriaga, A. Rodríguez-León, G. E. Ovando-Chacon, R. Salazar-Hernández, O. Montaña-Rivas y J. M. Canseco-Cortinas, "Medical Assistant Mobile Application for Diabetes Control by Simulating a Compartmental Model," *Applied Sciences*, vol. 10, n.º 19, oct. de 2020, <https://doi.org/10.3390/app10196846>, ISSN: 2076-3417. DOI: [10.3390/app10196846](https://doi.org/10.3390/app10196846). dirección: <https://www.mdpi.com/2076-3417/10/19/6846>.
- [8] **M. A. Nuño-Maganda**, C. Torres-Huitzil, Y. Hernández-Mier, J. De La Calleja, C. C. Martinez-Gil, J. H. B. Zambrano y A. D. Manríquez, "Smartphone-Based Remote Monitoring Tool for e-Learning," *IEEE Access*, vol. 8, págs. 121409-121423, jun. de 2020, <https://doi.org/10.1109/ACCESS.2020.3005330> ISSN: **2169-3536**. DOI: [10.1109/ACCESS.2020.3005330](https://doi.org/10.1109/ACCESS.2020.3005330). dirección: [10.1109/ACCESS.2020.3005330](https://doi.org/10.1109/ACCESS.2020.3005330).
- [7] Y. Hernández-Mier, **M. A. Nuño-Maganda**, S. Polanco-Martagón y M. del Refugio García Chávez, "Machine Learning Classifiers Evaluation for Automatic Karyogram Generation from G-Banded Metaphase Images," *Applied Sciences*, vol. 10, n.º 8, abr. de 2020, Article ID: **applsci-759087**, ISSN: **2076-3417**. DOI: [10.3390/app10082758](https://doi.org/10.3390/app10082758).
- [6] M. Hernández-Ordoñez, **M. A. Nuño-Maganda**, C. A. Calles-Arriaga, O. Montaña-Rivas y K. E. B. Hernández, "An Education Application for Teaching Robot Arm Manipulator Concepts Using Augmented Reality," *Mobile Information Systems*, vol. 2018, ago. de 2018, <https://doi.org/10.1155/2018/6047034>, Article ID: **6047034**, ISSN: **1875-905X**.
- [5] **M. A. Nuño-Maganda**, H. Herrera-Rivas, C. Torres-Huitzil, H. Marisol Marín-Castro e Y. Coronado-Pérez, "On-Device Learning of Indoor Location for WiFi Fingerprint Approach," *Sensors*, vol. 18, n.º 7, jul. de 2018, <https://doi.org/10.3390/s18072202>, Article ID: **2202**, ISSN: **1424-8220**.
- [4] S. Y. Carrizales-Villagómez, **M. A. Nuño-Maganda** y J. Rubio-Loyola, "A Platform for e-Health Control and Location Services for Wandering Patients," *Mobile Information Systems*, vol. 18, abr. de 2018, <https://doi.org/10.1155/2018/8164376>, Article ID: **8164376**, ISSN: **1875-905X**.
- [3] H. M. Marín-Castro, V. Sosa y **M. A. Nuño-Maganda**, "Automatic construction of vertical search tools for the Deep Web," *IEEE Latin America Transactions*, vol. 16, n.º 2, págs. 574-584, feb. de 2018, <https://doi.org/10.1109/TLA.2018.8327415>, ISSN: **1548-0992**.
- [2] M. Delgadillo-Escobar, C. Torres-Huitzil y **M. A. Nuño-Maganda**, "Comparison between 2D cellular automata based pseudorandom number generators," *IEICE Electronics Express*, vol. 9, n.º 17, págs. 1391-1396, sep. de 2012, <https://doi.org/10.1587/elex.9.1391>, ISSN: **1349-2543**, Indexado SCI.

- [1] C. Polanco-González, **M. A. Nuño-Maganda**, M. Arias-Estrada y G. del Río, “An FPGA Implementation to Detect Selective Cationic Antibacterial Peptides,” *PLoS ONE*, vol. 6, n.º 6, e21399, jun. de 2011, <https://doi.org/10.1371/journal.pone.0021399>, **ISSN: 1932-6203**, **Indexado SCI**. DOI: [10.1371/journal.pone.0021399](https://doi.org/10.1371/journal.pone.0021399).